

FindAndOpen

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

Someone might have hidden the password in the trace file.

Find the key to unlock this file. This tracefile might be good to analyze.

Hints

- Download the pcap and look for the password or flag.
 - Don't try to use a password cracking tool, there are easier ways here.
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Tools Used

- `file` – Identify PCAP file type
 - **Wireshark** – GUI tool for analyzing network traffic
 - `tshark` – Command-line Wireshark tool for packet analysis
 - `sort` – Organize timestamps in chronological order
 - `cut` – Extract specific fields from output
 - `xxd` – Convert hexadecimal to binary data
 - `base64` – Decode Base64-encoded content
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Complete Solution

Step 1: Download and Prepare

We have two files to download:

- `flag.zip` – An encrypted ZIP archive that requires a password
- `dump.pcap` – A network packet capture that likely contains the password

Step 2: Basic File Inspection

First, verify the file type and general information:

```
file dump.pcap
```

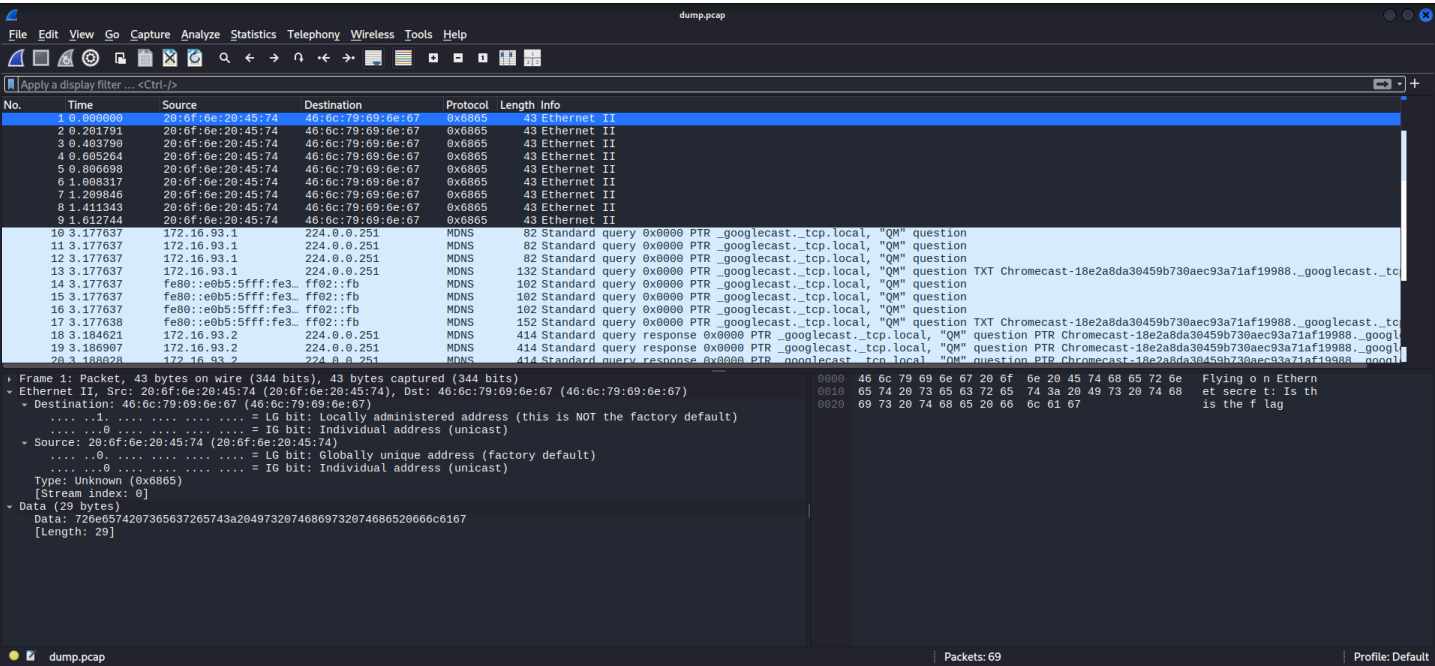
Expected output:

```
dump.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet,
capture length 262144)
```

The file is a network packet capture (PCAP) that requires analysis using network forensics tools.

Step 3: Initial Wireshark Analysis

Open the PCAP file in Wireshark to get a visual understanding of the traffic:



Scrolling through the capture, we notice an interesting message early on:

```
Flying on Ethernet secret: Is this the flag
```

This suggests that the password or flag might be hidden in the Ethernet layer.

Step 4: Examine Ethernet Traffic with tshark

Let's list all packets in the capture:

```
tshark -r dump.pcap
```

Output (excerpt):

```
1  0.000000 20:6f:6e:20:45:74 → 46:6c:79:69:6e:67 0x6865 43 Ethernet II
2  0.201791 20:6f:6e:20:45:74 → 46:6c:79:69:6e:67 0x6865 43 Ethernet II
3  0.403790 20:6f:6e:20:45:74 → 46:6c:79:69:6e:67 0x6865 43 Ethernet II
4  0.605264 20:6f:6e:20:45:74 → 46:6c:79:69:6e:67 0x6865 43 Ethernet II
5  0.806698 20:6f:6e:20:45:74 → 46:6c:79:69:6e:67 0x6865 43 Ethernet II
6  1.008317 20:6f:6e:20:45:74 → 46:6c:79:69:6e:67 0x6865 43 Ethernet II
7  1.209846 20:6f:6e:20:45:74 → 46:6c:79:69:6e:67 0x6865 43 Ethernet II
8  1.411343 20:6f:6e:20:45:74 → 46:6c:79:69:6e:67 0x6865 43 Ethernet II
9  1.612744 20:6f:6e:20:45:74 → 46:6c:79:69:6e:67 0x6865 43 Ethernet II
10 3.177637 172.16.93.1 → 224.0.0.251 MDNS 82 Standard query 0x0000 PTR
    _googlecast._tcp.local, "QM" question
```

We see many Ethernet II packets with a specific EtherType value (`0x6865`).

Step 5: Filter Suspicious Ethernet Type

After investigation, we notice a more interesting EtherType: `0x4c4b` (which translates to "LK" in ASCII – possibly "Lock" or a custom protocol).

Let's filter packets with this EtherType:

```
tshark -r dump.pcap -Y "eth.type == 0x4c4b" -T fields -e frame.time -e data.data
```

Output:

```
2022-10-22T09:33:33.157638000+0200
564768706379427063794230614755676332566a636d56304f69427761574e76513152476531497a4e4552
```

4a546b64665445394c5a46383d

We have a hexadecimal string in the `data.data` field.

Step 6: Decode the Extracted Data

The data appears to be encoded in multiple layers. Let's decode it step by step:

```
tshark -r dump.pcap -Y "eth.type == 0x4c4b" -T fields -e frame.time -e data.data |  
sort | cut -f2 -d $'\t' | xxd -r -p | base64 -d
```

Output:

This is the secret: <REDACTED>

Key finding: The extracted secret is the password for the encrypted ZIP file!

Step 7: Unlock the ZIP Archive

Now that we have the password, let's extract the `flag.zip` archive:

```
7z x flag.zip
```

Output:

```
7-Zip 26.00 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-02-12  
64-bit locale=C.UTF-8 Threads:4 OPEN_MAX:1024, ASM
```

```
Scanning the drive for archives:  
1 file, 231 bytes (1 KiB)
```

```
Extracting archive: flag.zip
```

```
--
```

```
Path = flag.zip
```

```
Type = zip
```

```
Physical Size = 231
```

Enter password (will not be echoed):

Enter the password we discovered in the PCAP file.

Step 8: Read the Flag

After successful extraction, a file named `flag` appears. Let's read it:

```
cat flag
```

Output:

```
picoCTF{<REDACTED>}
```



Final Result

```
picoCTF{<REDACTED>}
```

Note: The actual flag and password are redacted here. In the real challenge, the decoding steps will reveal the complete password and flag.



Key Concepts

Concept	Description
PCAP Analysis	Network packet captures contain timestamped records of all traffic, useful for forensic investigations.
EtherType	A two-byte field in an Ethernet frame that indicates which protocol is encapsulated in the payload.

Concept	Description
Custom EtherType	Values like <code>0x4c4b</code> may indicate custom or proprietary protocols used to hide data.
Packet Filtering	Using display filters (<code>-Y</code> option in tshark) to focus on specific traffic patterns.
Payload Extraction	Recovering data hidden in Ethernet payloads using field extraction and hex conversion.
Encoding Layers	Data can be encoded in multiple formats (hex → binary → Base64) requiring sequential decoding.
ZIP Password Recovery	Instead of brute-forcing, the password can be found elsewhere in the network capture.

Pro Tips

1. **Check all protocol layers** – Don't just look at TCP/UDP. Data can be hidden in Ethernet, ARP, ICMP, or other lower-level protocols.
2. **Look for unusual EtherType values** – Standard values are `0x0800` (IPv4), `0x0806` (ARP), `0x86DD` (IPv6). Unusual values often indicate hidden data.
3. **Combine multiple packets** – Use `sort` with timestamps to reconstruct data spread across multiple packets.
4. **Don't brute-force ZIP passwords** – The hint explicitly says there are easier ways. Always look for the password in associated files or network captures.
5. **Use `xxd -r -p`** – This converts hexadecimal strings back to raw binary data.